```
switch x^2 + y^2
  case 0
    label ← “origin”
  case [0,1]
    label ← “inside unit circle”
  case {1..10}
    label ← “multiple of unit circle”
  default
    label ← “outside unit circle”
```

Above, we use four different rule types for matching:

- the first one matches exactly one value (`0`)
- the second one matches all numbers between `0` and `1` included
- the third rule matches all integers between `1` and `10` included
- the last one matches all values (by default)

Notice that rules are evaluated from top to bottom and the first clause matching will be selected, so for instance while the value `0` for $x^2 + y^2$ would match both the first and second (and fourth!) clause, only the top-most one will match.

Finally, we have shown examples evaluating numeric expressions, but we will allow all kind of values (for instance strings) to be switched on.

# A.4 Loops